

Amir Bar

INFORMATION	Assistant Professor Imperial College London Office 407C, Huxley Building London, UK	Email: amirb4r@gmail.com Homepage: amirbar.net Google Scholar: [link] GitHub: [link]
EDUCATION	Tel Aviv University , Tel Aviv, Israel <i>Doctor of Philosophy</i> , Computer Science Dissertation: From Task-Specific to General Self-Supervised Machine Vision Advisors: Amir Globerson and Trevor Darrell (UC Berkeley).	July 2020 - Aug 2024
	UC Berkeley , Berkeley, California Visiting PhD Student	Aug 2021 - Aug 2024
	Tel Aviv University , Tel Aviv, Israel <i>Master of Science</i> , Computer Science Advisor: Lior Wolf.	Oct 2015 - July 2017
	Tel Aviv University , Tel Aviv, Israel <i>Bachelor of Science</i> , Computer Science <i>Magna Cum Laude</i>	Oct 2012 - July 2015
WORK EXPERIENCE	Imperial College London: Assistant Professor Building a new group with focus on Physical AI.	June 2026 - Now
	AMI Labs: Founding Member World modeling and planning.	June 2026 - Now
	Meta AI Research: Research Scientist Self-supervised learning and world modeling.	Aug 2025 - June 2026
	Meta AI Research: Postdoctoral Researcher Working on self-supervised learning and world modeling. Host: Professor Yann LeCun, Meta VP and an ACM Turing award laureate.	Aug 2024 - Aug 2025
	Zebra Medical Vision: ML Team leader Automated the reading of CT scans Four algorithms clinically approved by the US FDA.	Oct 16 - June 22
INTERESTS	World Modeling, Self-Supervised Learning, Planning	
CONFERENCE PUBLICATIONS	Alex N. Wang, Trevor Darrell, Pavel Izmailov, Yutong Bai, Amir Bar . “Lifting Embodied World Models for Planning and Control.” <i>In Proceedings of the European Conference on Computer Vision (ECCV)</i> . 2026.	
	Lorenzo Mur-Labadia, Matthew Muckley, Amir Bar , Mido Assran, Koustuv Sinha, Mike Rabbat, Yann LeCun, Nicolas Ballas, Adrien Bardes. “V-JEPA 2.1: Unlocking Dense Features in Video Self-Supervised Learning.” <i>In Proceedings of the European Conference on Computer Vision (ECCV)</i> . 2026.	
	Raktim Goswami, Amir Bar , David Fan, Tsung-Yen Yang, Gaoyue Zhou, Prashanth Krishnamurthy, Michael Rabbat, Farshad Khorrami, Yann LeCun. “World Models Can Leverage Human Videos for Dexterous Manipulation.” <i>In Proceedings of the European Conference on Computer Vision (ECCV)</i> . 2026.	
	Michael Psenka, Michael Rabbat, Aditi Krishnapriyan, Yann LeCun, Amir Bar . “Parallel Stochastic Gradient-Based Planning for World Models.” <i>Proceedings of the</i>	

43rd International Conference on Machine Learning (ICML). 2026.

Shengbang Tong, David Fan, John Nguyen, Ellis L. Brown II, Gaoyue Zhou, Shengyi Qian, Boyang Zheng, Théophane Vallaey, Junlin Han, Rob Fergus, Naila Murray, Marjan Ghazvininejad, Mike Lewis, Nicolas Ballas, **Amir Bar**, Michael Rabbat, Jakob Verbeek, Luke Zettlemoyer, Koustuv Sinha, Yann LeCun, Saining Xie. “Beyond Language Modeling: An Exploration of Multimodal Pretraining.” *Proceedings of the 43rd International Conference on Machine Learning (ICML)*. 2026. **Spotlight**.

Rim Assouel, **Amir Bar**, Michal Drozdal, Adriana Romero-Soriano. “PGT: Procedurally Generated Tasks for Improving Visual Grounding in MLLMs.” *Proceedings of the 43rd International Conference on Machine Learning (ICML)*. 2026.

Basile Terver, Randall Balestrieri, Megi Dervishi, David Fan, Quentin Garrido, Tushar Nagarajan, Koustuv Sinha, Wancong Zhang, Mike Rabbat, Yann LeCun*, **Amir Bar***. “A Lightweight Library for Energy-Based Joint-Embedding Predictive Architectures.” *ICLR World Models Workshop*. 2026. **Outstanding Paper Award**.

James Ni*, Zekai Wang*, Wei Lin*, **Amir Bar***, Trevor Darrell[†], Yann LeCun[†], Jitendra Malik[†], Roei Herzig[†]. “From Generated Human Videos to Physically Plausible Robot Trajectories.” *In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2026. **Findings**.

Xiaochuang Han, Youssef Emad, Melissa Hall, John Nguyen, Karthik Padthe, Liam Robbins, **Amir Bar**, DeLong Chen, Michal Drozdal, Maha Elbayad, Yushi Hu, Shangwen Li, Sreya Dutta Roy, Jakob Verbeek, XuDong Wang, Marjan Ghazvininejad, Luke Zettlemoyer, Emily Dinan. “TV2TV: A Unified Framework for Interleaved Language and Video Generation.” *In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2026.

Karen Ullrich, Jingtong Su, Claudia Shi, Arjun Subramonian, **Amir Bar**, Ivan Evtimov, Nikolaos Tsilivis, Randall Balestrieri, Julia Kempe, Mark Ibrahim. “OpenApps: Simulating Environment Variations to Measure UI-Agent Reliability.” *In International Conference on Learning Representations (ICLR)*. 2026. **Oral**.

Yutong Bai*, Danny Tran*, **Amir Bar***, Yann LeCun[†], Trevor Darrell[†], Jitendra Malik[†]. “Whole-Body Conditioned Egocentric Video Prediction.” *In Advances in Neural Information Processing Systems (NeurIPS)*. 2025.

Michal Golovanevsky*, William Rudman*, Michael Lepori, **Amir Bar**, Ritambhara Singh, Carsten Eickhoff. “Pixels Versus Priors: Controlling Knowledge Priors in Vision-Language Models through Visual Counterfactuals.” *In Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2025.

David Fan*, Shengbang Tong*, Jiachen Zhu, Koustuv Sinha, Zhuang Liu, Xinlei Chen, Michael Rabbat, Nicolas Ballas, Yann LeCun, **Amir Bar**[†], Saining Xie[†]. “Scaling Language-Free Visual Representation Learning.” *In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 2025. **Spotlight**.

William Rudman, Michal Golovanevsky, **Amir Bar**, Vedant Palit, Yann LeCun, Carsten Eickhoff, Ritambhara Singh. “Forgotten Polygons: Multimodal Large Language Models are Shape-Blind.” *The 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*. 2025.

Grace Luo, Trevor Darrell, **Amir Bar**. “Vision-Language Models Create Cross-Modal Task Representations.” *Proceedings of the 42th International Conference on Machine Learning (ICML)*. 2025.

Amir Bar, Gaoyue Zhou, Danny Tran, Trevor Darrell, Yann LeCun. “Navigation World Models.” *In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2025. **Best Paper, Honorable Mention**.

Alberto Hojel, Yutong Bai, Trevor Darrell, Amir Globerson, **Amir Bar**. “Finding Visual Task Vectors”. *In European Conference on Computer Vision (ECCV)*. 2024.

Amir Bar, Arya Bakhtiar, Antonio Loquercio, Jathushan Rajasegaran, Danny Tran, Yann LeCun, Amir Globerson, Trevor Darrell. “EgoPet: A pet’s-eye view of the world for learning animal behavior”. *In European Conference on Computer Vision (ECCV)*. 2024.

Amir Bar, Florian Bordes, Assaf Shocher, Mahmoud Assran, Pascal Vincent, Nicolas Ballas, Trevor Darrell, Amir Globerson, and Yann LeCun. “Stochastic positional embeddings improve masked image modeling.” *Proceedings of the 41th International Conference on Machine Learning (ICML)* 2024.

Yutong Bai, Xinyang Geng, Karttikeya Mangalam, **Amir Bar**, Alan Yuille, Trevor Darrell, Jitendra Malik, Alexei A Efros. “Sequential Modeling Enables Scalable Learning for Large Vision Models”. *In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2024.

Amir Bar*, Yossi Gandelsman*, Trevor Darrell, Amir Globerson, Alexei Efros. “Visual Prompting via Image Inpainting.” *In Advances in Neural Information Processing Systems (NeurIPS)*. 2022.

Amir Bar, Xin Wang, Vadim Kantorov, Colorado J Reed, Roei Herzig, Gal Chechik, Anna Rohrbach, Trevor Darrell, and Amir Globerson. “DETReg: Unsupervised Pre-training with Region Priors for Object Detection.” *In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2022.

Elad Ben-Avraham, Roei Herzig, Karttikeya Mangalam, **Amir Bar**, Anna Rohrbach, Leonid Karlinsky, Trevor Darrell, Amir Globerson. “Bringing Image Scene Structure to Video via Frame-Clip Consistency of Object Tokens.” *In Advances in Neural Information Processing Systems (NeurIPS)*. 2022.

Roei Herzig, Elad Ben-Avraham, Karttikeya Mangalam, **Amir Bar**, Gal Chechik, Anna Rohrbach, Trevor Darrell, Amir Globerson. “Object-Region Video Transformers.” *In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2022.

Raouf Muhamedrahimov, **Amir Bar**, and Ayelet Akselrod-Ballin. “Learning Interclass Relations for Intravenous Contrast Phase Classification in CT.” *In Medical Imaging with Deep Learning (MIDL)*. 2021.

Amir Bar*, Herzig, Roei*, Xiaolong Wang, Anna Rohrbach, Gal Chechik, Trevor Darrell, and Amir Globerson. “Compositional Video Synthesis with Action Graphs.” *Proceedings of the 38th International Conference on Machine Learning (ICML)* 2021.

Herzig, Roei*, **Amir Bar***, Huijuan Xu, Gal Chechik, Trevor Darrell, and Amir Globerson. “Learning Canonical Representations for Scene Graph to Image Generation.” *In European Conference on Computer Vision (ECCV)*. 2020.

David Chettrit, Tomer Meir, Hila Lebel, Mila Orlovsky, Ronen Gordon, Ayelet Akselrod-Ballin, **Amir Bar**. “3D Convolutional Sequence to Sequence Model for Vertebral Compression Fractures Identification in CT.” *In Medical Image Computing and Computer Assisted Intervention (MICCAI)*. 2020.

Ginosar Shiry*, **Amir Bar***, Gefen Kohavi, Caroline Chan, Andrew Owens, and Jitendra Malik. “Learning Individual Styles of Conversational Gesture.” *In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2019.

Amir Bar, Michal Mauda Havakuk, Yoni Turner, Michal Safadi, and Eldad Elnekave. “Improved ich classification using task-dependent learning.” *In IEEE 16th International Symposium on Biomedical Imaging (ISBI)*. 2019.

Ofir Press*, **Amir Bar***, Ben Bogin*, Jonathan Berant and Lior Wolf. “Language generation with recurrent generative adversarial networks without pre-training.” *In the 1st Workshop on Learning to Generate Natural Language (ICMLW)*. 2017.

Amir Bar, Lior Wolf, Orna Bergman Amitai, Eyal Toledano and Eldad Elnekave. “Compression fractures detection on CT.” *In Proceedings of SPIE Medical Imaging*. 2017.

JOURNAL PUBLICATIONS

Jiarui Xu, Yossi Gandelsman, **Amir Bar**, Jianwei Yang, Jianfeng Gao, Trevor Darrell, Xiaolong Wang. “IMProv: Inpainting-based Multimodal Prompting for Computer Vision Tasks”. *Transactions on Machine Learning Research (TMLR)*. 2024.

Muhamedrahimov, Raouf, **Amir Bar**, Jonathan Laserson, Ayelet Akselrod-Ballin, and Eldad Elnekave. “Using machine learning to identify intravenous contrast phases on computed tomography.” *In Computer Methods and Programs in Biomedicine 215*. 2022.

Noa Dagan, Eldad Elnekave, Noam Barda, Orna Bergman-Amitai, **Amir Bar**, Mila Orlovsky, Eitan Bachmat, Ran D. Balicer. “Automated opportunistic osteoporotic fracture risk assessment using computed-tomography scans to aid in FRAX underutilization”. *In Nature Medicine*. 2020.

Krishnaraj, Arun, Spencer Barrett, Orna Bregman-Amitai, Michael Cohen-Sfady, **Amir Bar**, David Chettrit, Mila Orlovsky, and Eldad Elnekave. “Simulating dual-energy X-ray absorptiometry in CT using deep-learning segmentation cascade.” *In Journal of the American College of Radiology*. 2019.

PATENTS

Ronen Gordon, **Amir Bar**, Raouf Muhamedrahimov, Ayelet Akselrod-Ballin. “Identifying calcification lesions in contrast enhanced images.”. *U.S. Patent*. 2024.

Amir Bar. “Systems and methods for automated detection of visual objects in medical images.”. *U.S. Patent*. 2023.

Amir Bar. “Identification of a contrast phase depicted in a medical image”. *U.S. Patent*. 2023.

Amir Bar, Raouf Muhamedrahimov, and Rachel Wities. “Cross modality training of machine learning models”. *U.S. Patent*. 2023.

INVITED TALKS

CVEU CVPR Workshop, June 2026
Princeton University, November 2025
Oxford University, September 2025
Meta (FAIR), June 2025
Imperial College London, March 2025
Johns Hopkins University, March 2025
Toyota Technological Institute at Chicago, February 2025
University of Washington, February 2025
Brown University, November 2024
Hebrew University of Jerusalem, July 2024

Weizmann Institute of Science, July 2024
Meta (FAIR), June 2024
NVIDIA AI, June 2024
ByteDance, June 2024
Berkeley AI Research, December 2023
Structured Representations for Video Understanding Workshop, ICCV 2021
Learning from Limited and Imperfect Data Workshop, CVPR 2021
Hebrew University of Jerusalem, 2019
Medical Machine Learning Israel, 2019

HONORS & AWARDS

Best Paper, Honorable Mention. CVPR 2025
Blavatnik Prize for Computer Science Doctoral Fellows, 2025
Winner of the Ego4D PNR Temporal Localization Challenge, CVPR 2022
Award for Outstanding Academic Achievements, TAU CS (top 5%). 2016
Magna Cum Laude, BSc in Computer Science, Tel Aviv University, 2015
Award for Outstanding Academic Achievements, TAU CS (top 5%). 2015

SERVICE

Organizing Committee:

- Robust and Interactable World Models Workshop at ICCV 2025
- Primary Organizer - First Workshop on Visual Prompting at CVPR 2024
- Assistant Program Chair - NeurIPS 2023

Area Chair: ICLR, NeurIPS.

Reviewer: CVPR, NeurIPS, ICML, ICCV, ECCV, TPAMI.

Admission Committee: UC Berkeley, 2023.